Question 1
Problem Statement:
Normalize the table up to 3NF.
Dataset:

OrderID	CustomerName	Product	Price
1	Anuj	Laptop	50000
1	Anuj	Mouse	500
2	Amit	Laptop	50000
2	Amit	Keyboard	1500
3	Ravi	Mouse	500
4	Sita	Laptop	50000
5	Mohan	Keyboard	1500
6	Riya	Mouse	500
7	Karan	Laptop	5000
8	Neha	Keyboard	1500

Answer1: Table1:Customer_Order

OrderID	CustomerName
1	Anuj
2	Amit
3	Ravi
4	Sita
5	Mohan
6	Riya
7	Karan
8	Neha

Table2:Order_Product

OrderID	Product
1	Laptop
1	Mouse
2	Laptop
2	Keyboard
3	Mouse
4	Laptop
5	Keyboard
6	Mouse

7	Laptop
8	Keyboard

Table3:Product

Product	Price
Laptop	50000
Mouse	500
Keyboard	1500

Question 2
Problem Statement:
Identify dependencies and normalize to 3NF.
Dataset:

EmpID	EmpName	DeptID	DeptName
101	Anuj	D1	HR
102	Amit	D2	IT
103	Ravi	D1	HR
104	Sita	D3	Finance
105	Mohan	D2	IT
106	Riya	D1	HR
107	Karan	D3	Finance
108	Neha	D2	IT
109	Pooja	D1	HR
110	Rahul	D3	Finance

Answer2:1. Identify Dependencies
Functional Dependencies:
EmpID→EmpName,DeptID
DeptID→DeptName

Table1:Employee

EmpID	EmpName	DeptID
101	Anuj	D1
102	Amit	D2
103	Ravi	D1
104	Sita	D3
105	Mohan	D2
106	Riya	D1
107	Karan	D3

108	Neha	D2
109	Pooja	D1
110	Rahul	D3

Table2:Department

DeptID	DeptName
D1	HR
D2	IT
D3	Finance

Question 3
Problem Statement:
Remove transitive dependency and normalize.
Dataset:

StudentID	Subject	Teacher	TeacherPhone
1	Math	Ravi	9999
1	Science	Ravi	9999
2	English	Sita	8888
3	Math	Ravi	9999
4	Science	Ravi	9999
5	English	Sita	8888
6	Math	Ravi	9999
7	Science	Ravi	9999
8	English	Sita	8888
9	Math	Ravi	9999

Answer:3 1.Functional Dependencies
Teacher → TeacherPhone
StudentID → Teacher
This is a Transitive Dependency

2.Remove Transitive Dependency
Table1:Student Details

StudentID	Subject	Teacher
1	Math	Ravi
1	Science	Ravi
2	English	Sita
3	Math	Ravi

4	Science	Ravi
5	English	Sita
6	Math	Ravi
7	Science	Ravi
8	English	Sita
9	Math	Ravi

Table2:Teacher Details

Teacher	TeacherPhone
Ravi	9999
Sita	8888

3.Now the tables are in 3NF.

Question 4
Problem Statement:
Normalize to 3NF and identify all dependencies.
Dataset:

BookID	Author	Publisher	PublisherAddress
B1	A1	P1	Delhi
B2	A2	P2	Mumbai
B3	A1	P1	Delhi
B4	A3	P3	Pune
B5	A2	P2	Mumbai
B6	A1	P1	Delhi
B7	A3	P3	Pune
B8	A2	P2	Mumbai
B9	A1	P1	Delhi
B10	A3	P3	Pune

Answer4:1.Normalisation to 3NF
1NF:Table is already in 1NF
2NF:Table is in 2NF
3NF:There is a transitive Dependency:Publisher →PublisherAddress
Decompose it:
Table1:Book Table

BookID	Author	Publisher
B1	A1	P1
B2	A2	P2

B3	A1	P1
B4	A3	P3
B5	A2	P2
B6	A1	P1
B7	A3	P3
B8	A2	P2
B9	A1	P1
B10	A3	P3

Table2:Publisher Table

Publisher	PublisherAddress
P1	Delhi
P2	Mumbai
P3	Pune

2.Identify all Dependencies
1.Functional Dependencies: BookID→Author,Publisher
Publisher→PublisherAddress
2.Transitive Dependency: BookID→Publisher→PublisherAddress

Question 5
Problem Statement:
Find candidate key and normalize to 3NF
Dataset:

OrderID	ProductID	ProductName	CustomerID	CustomerName
1	P1	Laptop	C1	Anuj
1	P2	Mouse	C1	Anuj
2	P1	Laptop	C2	Amit
2	P3	Keyboard	C2	Amit
3	P2	Mouse	C3	Ravi
4	P1	Laptop	C4	Sita
5	P3	Keyboard	C5	Mohan
6	P2	Mouse	C6	Riya
7	P1	Laptop	C7	Karan
8	P3	Keyboard	C8	Neha

Answer5:1. Candidate Key=(OrderID,ProductID)
2. Normalisation:1NF:Already Satisfied
2NF:Remove partial dependency
Create:

Table1:Order Details

OrderID	ProductID
1	P1
1	P2
2	P1
2	P3
3	P2
4	P1
5	P3
6	P2
7	P1
8	P3

Table2:Product Table

ProductID	ProductName
P1	Laptop
P2	Mouse
P3	Keyboard

Table3:Customer Table

CustomerID	CustomerName
C1	Anuj
C2	Amit
C3	Ravi
C4	Sita
C5	Mohan
C6	Riya
C7	Karan
C8	Neha

Table4:Order-Customer Table

OrderID	CustomerID
1	C1
2	C2
3	C3
4	C4

5	C5
6	C6
7	C7
8	C8

3.Relation is normalised to 3NF

Question 6
Problem Statement:
Normalize the student-course table up to 3NF.
Dataset:

StudentID	StudentName	Course	Instructor
1	Anuj	Math	Ravi
1	Anuj	Science	Ravi
2	Amit	English	Sita
3	Ravi	Math	Ravi
4	Sita	Science	Ravi
5	Mohan	English	Sita
6	Riya	Math	Ravi
7	Karan	Science	Ravi
8	Neha	English	Sita
9	Pooja	Math	Ravi

Answer6:1.Functional Dependencies: StudentID→StudentName
Course→Instructor
2. Relation is in 1NF
Partial dependency exists: StudentID→StudentName in 2NF
Decompose:
Table1:Student Table

StudentID	StudentName
1	Anuj
2	Amit
3	Ravi
4	Sita
5	Mohan
6	Riya
7	Karan
8	Neha

9	Pooja

Table2: Student-Course table

StudentID	Course
1	Math
1	Science
2	English
3	Math
4	Science
5	English
6	Math
7	Science
8	English
9	Math

3.3NF:Transitive Dependency Exists:
Course→Instructor
Decompose:
Table3:

Course	Instructor
Math	Ravi
Science	Ravi
English	Sita

Question 7
Problem Statement:
Normalize employee-project data to 3NF.
Dataset:

EmpID	ProjectID	ProjectName	Manager
101	P1	Alpha	Raj
101	P2	Beta	Raj
102	P1	Alpha	Raj
103	P3	Gamma	Simran
104	P2	Beta	Raj
105	P3	Gamma	Simran
106	P1	Alpha	Raj
107	P2	Beta	Raj
108	P3	Gamma	Simran

109	P1	Alpha	Raj

Answer7:1.Functional Dependencies: ProjectID→ProjectName
ProjectID→Manager

2.Relation is in 1NF
Partial Dependency exists: ProjectID→ProjectName,Manager

Decompose:

Table1:Employee-Project Table

EmpID	ProjectID
101	P1
101	P2
102	P1
103	P3
104	P2
105	P3
106	P1
107	P2
108	P3
109	P1

Table2:Project Table

ProjectID	ProjectName	Manager
P1	Alpha	Raj
P2	Beta	Raj
P3	Gamma	Simran

3. No transitive dependency exists.
Relation is normalised to 3NF.

Question 8

Problem Statement:

Normalize sales data up to 3NF

Dataset:

SaleID	Product	Category	Price
1	Laptop	Electronics	50000
2	Mouse	Electronics	500
3	Keyboard	Electronics	1500
4	Shirt	Clothing	1000
5	Jeans	Clothing	2000
6	Laptop	Electronics	50000

7	Mouse	Electronics	500
8	Shirt	Clothing	1000
9	Jeans	Clothing	2000
10	Keyboard	Electronics	1500

Answer8:1.Functional Dependencies: Product→Category
Product→Price

2. Relation is in 1NF
Relation is in 2NF
Transitive dependency exists in 3NF: SalesID→Product→Category,Price

Decompose:

Table1:Sales Table

SaleID	Product
1	Laptop
2	Mouse
3	Keyboard
4	Shirt
5	Jeans
6	Laptop
7	Mouse
8	Shirt
9	Jeans
10	Keyboard

Table2: Product Table

Product	Category	Price
Laptop	Electronics	50000
Mouse	Electronics	500
Keyboard	Electronics	1500
Shirt	Clothing	1000
Jeans	Clothing	2000

Question 9

Problem Statement:

Normalize hospital patient records to 3NF.

Dataset:

PatientID	PatientName	Doctor	DoctorSpecialization
1	Anuj	Dr.Ravi	Cardiology
2	Amit	Dr.Sita	Neurology

3	Ravi	Dr.Ravi	Cardiology
4	Sita	Dr.Sita	Neurology
5	Mohan	Dr.Ravi	Cardiology
6	Riya	Dr.Sita	Neurology
7	Karan	Dr.Ravi	Cardiology
8	Neha	Dr.Sita	Neurology
9	Pooja	Dr.Ravi	Cardiology
10	Rahul	Dr.Sita	Neurology

Answer9:1.Functional Dependencies : PatientID→PatientName
Doctor→DoctorSpecialisation

2.Relation is in 1NF
Relation is in 2NF
Transitive dependency exists in 3NF: PatientID→Doctor→DoctorSpecialization

Decompose:

Table1: Patient Table

PatientID	PatientName	Doctor
1	Anuj	Dr.Ravi
2	Amit	Dr.Sita
3	Ravi	Dr.Ravi
4	Sita	Dr.Sita
5	Mohan	Dr.Ravi
6	Riya	Dr.Sita
7	Karan	Dr.Ravi
8	Neha	Dr.Sita
9	Pooja	Dr.Ravi
10	Rahul	Dr.Sita

Table2: Doctor Table

Doctor	DoctorSpecialization
Dr.Ravi	Cardiology
Dr.Sita	Neurology

Question 10

Problem Statement:

Normalize university data to 3NF

Dataset:

StudentID	Dept	HOD
1	CSE	Dr.A

2	IT	Dr.B
3	CSE	Dr.A
4	ECE	Dr.C
5	IT	Dr.B
6	CSE	Dr.A
7	ECE	Dr.C
8	IT	Dr.B
9	CSE	Dr.A
10	ECE	Dr.C

Answer10:1.Functional Dependencies:

StudentID→Dept

Dept→HOD

2. 1NF:Relation is in 1NF

2NF: No partial dependency exists

Relation is in 2NF

3NF:Transitive dependency exists:

StudentID→Depty→HOD

Decompose:

Table1:Student Table

StudentID	Dept
1	CSE
2	IT
3	CSE
4	ECE
5	IT
6	CSE
7	ECE
8	IT
9	CSE
10	ECE

Table2: Department Table

Dept	HOD
CSE	Dr.A
IT	Dr.B
ECE	Dr.C

3.The relation is normalized to 3NF.